

GAA Nanosheet FET TCAD - AI - PPA

Expanded Literature, Parameter-Range and Research-Method Guide

Purpose: This 10-page guide is written for team and supervisor discussions. It explains what previous GAA/NSFET papers already cover, why our candidate input parameters are being considered, how the working ranges should be justified, where overlap is acceptable, and what must be defined before claiming Power-Performance-Area optimization.

Core workflow: public industry/experimental reference -> calibrated TCAD baseline -> physically justified design space -> DOE/sampling -> TCAD dataset -> AI surrogate -> explicit PPA objectives -> Pareto optimization -> fresh TCAD re-validation.

Central rule: numerical ranges are not the novelty by themselves. Standard GAA dimensions will naturally overlap with previous work. The potential research strength comes from the engineering question, technology-constrained design space, interaction among variables, PPA formulation, optimization, validation and physical interpretation.

Provisional candidate inputs: Lg, WNS, TNS, Lsp, Tsp, NNS, EOT and VDD. S/D doping, channel doping, metal work function and materials are initially candidates to remain fixed at the calibrated reference.

Before simulation: the team must still lock the reference device and the exact mathematical definitions of Power, Performance and Area.

1. What Relevant Previous Papers Already Cover

Xu et al., IEEE Transactions on Electron Devices, 2022.

This is one of the most important overlap papers for our project. Xu and co-authors did not merely perform a conventional TCAD sweep. Their work combined a three-dimensional stacked GAA nanosheet transistor model with an artificial neural network and multi-objective optimization. They studied channel geometry together with the source/drain doping profile, then optimized several electrical characteristics. Therefore, we cannot claim that combining TCAD, machine learning, GAA nanosheets and multi-objective optimization is itself new.

The lesson for us is that changing the ANN model or slightly changing geometric ranges would not create a strong distinction. Our research question must go beyond the generic idea of 'AI optimizes a GAAFET.' The possible distinction we are investigating is explicit Power-Performance-Area optimization, technology-constrained inputs, stacking variables, operating voltage and TCAD re-validation of Pareto-selected devices.

Zhao et al., Micromachines, 2024.

This paper is important because its nominal nanosheet device is tied to an advanced-node reference and the TCAD model is calibrated against experimental DC and AC characteristics before the ANN dataset is generated. The reported nominal device uses approximately $L_g = 14$ nm, nanosheet width = 15 nm, nanosheet thickness = 6 nm, spacer length = 6 nm, sheet spacing = 10 nm, EOT = 1.35 nm, source/drain doping = $5 \times 10^{20} \text{ cm}^{-3}$ and metal work function = 4.4 eV.

For ANN modelling, the paper varies major geometry quantities including L_g over approximately 10-20 nm, width over 15-30 nm and thickness over 4-7 nm. This tells us two things. First, these dimensions are already established ML inputs, so using L_g , WNS and TNS is not a novelty claim. Second, a calibrated advanced-node baseline is much more defensible than selecting arbitrary ranges.

1. What Relevant Previous Papers Already Cover - Continued

Pawinanto et al., 2026.

This recent work comes very close to the simplest version of a TCAD-to-ML nanosheet project. It identifies five essential device factors: gate length, nanosheet width, nanosheet height/thickness, spacer length and EOT. The reported discrete design space includes gate length around 11-13 nm, sheet width around 21-29 nm, sheet height around 4-6 nm, spacer length around 3-5 nm and EOT around 1-2 nm. Hundreds of TCAD configurations are used to train an ANN for device-characteristic prediction.

This is why a project whose entire contribution is 'we varied Lg, WNS, TNS, Lsp and EOT, generated TCAD data and trained an ANN' would have a serious novelty problem. Those variables remain useful, but they need to serve a different engineering objective.

Representative geometry and device-physics studies.

Earlier GAA nanosheet studies already examine broad geometric regions. Representative work studies sheet widths extending roughly from 10 nm to 50 nm and sheet thicknesses around 5-9 nm, while other advanced-node studies cover gate lengths down toward the 8-20 nm region. These papers establish the familiar trade-offs: larger effective channel dimensions can increase drive capability, while thinner sheets and longer gates can improve electrostatic control; geometry also affects capacitance, RF response, leakage and circuit-related metrics.

Therefore, there is no scientifically useful strategy of searching for an 'unused' width or thickness interval. A range outside previous papers can even be less credible if it has no technology justification.

Overall literature conclusion.

The conventional geometry space is already occupied: Lg, WNS, TNS, spacer dimensions and EOT are standard research variables. Doping-profile optimization has also been combined with AI. ANN-based device prediction is established. Multi-objective optimization is established. Our paper must therefore be built around a more specific problem.

Candidate research question: Can a calibrated, industry/experiment-informed stacked GAA nanosheet design space be co-optimized for explicit Power, Performance and Area objectives by an AI surrogate, while satisfying electrostatic constraints and reproducing the selected Pareto designs in fresh TCAD simulations?

2. How We Should Choose the Working Ranges

The range-selection logic should have four stages. **First**, choose a credible experimental or industry-relevant reference for which enough public geometry and electrical data exist. **Second**, calibrate the TCAD model against that reference. **Third**, define a physically feasible variation around the calibrated device. **Fourth**, remove combinations that violate process, geometry or electrical constraints before training the AI model.

This is different from selecting a range because another paper did not use it. For example, if a paper stops at WNS = 50 nm, choosing 51-65 nm only to look different is not a meaningful contribution. A reviewer can ask why an advanced-node device should operate in that region.

Provisional working set:

Lg = 10-16 nm; WNS = 15-35 nm; TNS = 4-7 nm; Lsp = 3-7 nm; Tsp = 8-12 nm provisionally; NNS = 2, 3 or 4; EOT = approximately 0.7-1.3 nm provisionally; VDD = 0.5-0.7 V.

These bounds intentionally overlap published ranges. The overlap is acceptable because these are physical variables, not copied research conclusions. What must be original is our experiment design, dataset generation, PPA formulation, optimization, analysis and conclusions.

Variables initially held fixed: S/D doping, channel doping, work function, materials and other process/model parameters should initially follow the calibrated baseline. This keeps the research question interpretable. If sensitivity analysis later shows one of them dominates PPA, the team can reconsider it.

Why not full factorial? Eight variables with only a few levels each can create tens of thousands of combinations. This motivates DOE or intelligent sampling. The AI collaboration should reduce expensive TCAD sampling and explore the larger continuous design space rather than simply fitting a model after brute-force simulation.

3. Explanation of Every Proposed Input Parameter

Gate length (Lg).

Lg is the physical gate-controlled channel length. It is one of the strongest scaling variables. Reducing Lg can reduce device dimensions and can support faster switching, but gate control becomes harder as source and drain fields interact more strongly with the channel. The resulting penalties can include higher leakage, larger DIBL and degraded subthreshold behaviour. Our provisional 10-16 nm interval deliberately spans an advanced-node region in which these competing effects should be visible.

Nanosheet width (WNS).

WNS is the lateral width of an individual sheet. Because the gate surrounds the sheet, changing width changes the available effective conduction width and therefore drive capability. Wider sheets can increase ION, but they can also increase gate-related capacitance and change electrostatics and physical footprint. The proposed 15-35 nm interval is not new; its purpose is to let the optimizer see current-versus-capacitance/area interactions.

Nanosheet thickness (TNS).

TNS is the vertical thickness of each nanosheet channel. Thin sheets generally improve gate control of the channel and can help suppress short-channel effects. However, making the sheet too thin reduces the channel cross-section and can introduce stronger confinement and mobility-related effects depending on the physical models used. The 4-7 nm interval is a credible advanced-node region already represented in literature, making it useful for calibration and comparison.

Spacer length (Lsp).

The spacer separates the gate edge from heavily doped source/drain regions and affects the underlap/extension region. Its value can influence parasitic source/drain resistance, fringing capacitance, electrostatic coupling and switching behaviour. Very short spacers can increase unwanted coupling; very long spacers can increase access resistance and consume physical length. This makes Lsp useful when performance is defined through delay rather than only ION.

3. Explanation of Every Proposed Input Parameter - Continued

Sheet-to-sheet spacing (Tsp).

Tsp is the vertical separation between stacked nanosheets. It matters because a GAA nanosheet transistor gains drive capability by stacking channels vertically. Spacing affects total stack height, gate formation around the sheets, parasitic coupling and packing density. Many compact-model studies hold spacing fixed; therefore we want to test whether a narrow variation around a calibrated reference actually influences our PPA definition. The proposed 8-12 nm interval is provisional and must be checked against the selected process/reference.

Number of nanosheets (NNS).

NNS is the number of vertically stacked conducting sheets. Adding sheets increases total effective channel width and can increase drive current without increasing lateral footprint in the same way as a wider planar device. But more sheets also increase total gate/channel capacitance, stack height and fabrication complexity. NNS = 2, 3 and 4 gives a discrete architectural variable that could create a clear performance-versus-capacitance/density trade-off. We should only retain it if our area/density metric responds meaningfully to stacking.

Equivalent oxide thickness (EOT).

EOT expresses the electrostatic strength of the gate dielectric stack as an equivalent SiO₂ thickness. Smaller EOT generally improves gate-to-channel control and can reduce electrostatic degradation, but it also increases gate capacitance and may raise leakage/reliability concerns depending on the physical dielectric stack. Because published advanced devices use different gate-stack assumptions, our provisional 0.7-1.3 nm range must be replaced by a range justified from the calibrated reference.

Supply voltage (VDD).

VDD is not a geometric parameter; it is the operating voltage. We include it because PPA is an operating-point problem as well as a geometry problem. Dynamic switching power is approximately proportional to $C \times VDD^2 \times f$, so voltage reduction can strongly reduce power. However, lower VDD also reduces overdrive and available current, which can increase delay. Co-optimizing VDD with geometry may therefore reveal operating points that a geometry-only optimization misses.

Why doping and work function are initially fixed.

Source/drain doping and metal work function strongly influence resistance, threshold voltage and leakage. Allowing them to vary immediately would increase dimensionality and could let the optimizer improve results mainly by shifting VT or contact/access behaviour. Since geometry plus doping has already been optimized using AI in previous work, fixing these quantities to the calibrated reference gives us a cleaner first research question.

4. From TCAD Inputs to Measurable Outputs

The AI model should not receive a collection of inputs without a physics plan. Each TCAD design point needs a consistent extraction procedure. At minimum, the team should consider ION, IOFF, threshold voltage, subthreshold swing, DIBL and relevant capacitances. Depending on the final PPA definition, total gate capacitance, intrinsic capacitance components, resistance or other parasitic quantities may also be required.

Electrical constraints are essential.

A mathematical optimizer will exploit whatever freedom we give it. A design may appear excellent for one objective while having unacceptable leakage or electrostatics. Therefore the Pareto problem should contain feasibility constraints, for example limits on IOFF, SS and DIBL. The exact limits must come from the chosen technology target or justified literature, not from arbitrary numbers.

Calibration before dataset generation.

Before generating hundreds of TCAD points, one baseline device should reproduce credible published characteristics as closely as the available data permit. Calibration may involve geometry, mobility/transport models, quantum corrections, recombination, high-field effects, series resistance and other physical-model choices. The team should document which parameters are calibrated and which are directly taken from the reference.

Dataset integrity.

All designs must use consistent extraction definitions. For example, ION and IOFF must be measured at explicitly defined VGS and VDS conditions; VT extraction must use one method throughout; capacitance must be evaluated at specified bias points. Without this consistency, the AI model can learn numerical inconsistencies rather than device physics.

DOE role.

Instead of a full factorial sweep, investigate a sampling strategy that covers the multidimensional space efficiently. The team should compare candidate DOE concepts such as Latin hypercube, space-filling designs or sequential/adaptive sampling and justify the final choice. The correct sample count should be established from surrogate-learning curves and validation error rather than guessed in advance.

5. Critical Definitions We Must Lock Before Claiming PPA

Power.

The word 'power' is ambiguous at transistor level. We must decide whether the objective represents dynamic switching power, static/leakage power, total power under a specified activity condition, energy per transition, or a clearly labelled proxy. For dynamic power, a familiar relation is $P_{\text{dynamic}} \approx \alpha \times C \times V_{\text{DD}}^2 \times f$, but every term requires a definition. What capacitance is C ? What switching activity α is assumed? Is frequency fixed, derived from delay, or omitted? Leakage power may be approximated using V_{DD} and leakage current, but the exact bias and current definition must be stated.

The project should avoid calling a single current value 'power' unless the conversion and operating assumptions are explicit. If only transistor-level TCAD data are available, it may be more rigorous to report an energy/delay proxy than to pretend we have full circuit power.

Performance.

Performance also needs an exact metric. I_{ON} is useful, but maximum I_{ON} is not identical to switching speed. A more physical transistor-level speed proxy can involve intrinsic delay, commonly related to $C \times V / I$, provided the capacitance and current are consistently defined. Alternatively, a circuit-level inverter delay would be stronger but requires a compact/circuit methodology beyond a single-device DC simulation. The team must choose the level of abstraction and remain consistent.

5. Critical Definitions We Must Lock - Continued

Area.

Area is the hardest term in our project. A vertically stacked nanosheet transistor does not map cleanly to a simple width x length expression. We must decide whether 'Area' means lateral transistor footprint, a contacted-gate/source-drain pitch proxy, a density-related metric, or an estimate connected to standard-cell area. Vertical stack height by itself is not silicon footprint, so we should not call it area without justification.

This decision directly determines whether Tsp and NNS belong in the optimizer. If the area metric ignores vertical stacking entirely, varying sheet spacing may not contribute to the Area objective. If we use a technology-density proxy that includes pitch/stack constraints, then the relationship needs to be derived and defended. This is a major research-design question, not a formatting detail.

Pareto optimality.

With multiple objectives there is usually no single 'best' device. A Pareto-optimal design is one for which improving one objective would worsen at least one other objective. The final paper should therefore show a set or frontier of non-dominated designs rather than simply announce one optimum. We should then select representative points - for example a low-power point, a high-performance point and a balanced point - and re-simulate them in TCAD.

Validation.

There are two different validations. First, the baseline TCAD model is validated/calibrated against published experimental/reference data. Second, the AI surrogate is validated against TCAD points that were not used for training. Finally, selected AI/Pareto designs should be run again in TCAD. These stages answer different questions and should not be mixed together.

Publication strength comes from agreement between the mathematical optimization and the physical TCAD behaviour, followed by an explanation of why the trade-offs occur.

6. Recommended Team Workflow, Novelty Boundary and Conclusion

Stage 1 - Build the literature matrix. Expand beyond the core papers and record device type, node/reference, geometry ranges, process parameters, TCAD models/calibration, ML inputs, ML outputs, optimization objectives and validation method.

Stage 2 - Select the reference. Compare public Samsung MBCFET-related information, Intel RibbonFET, TSMC nanosheet disclosures, imec demonstrations and IEDM/VLSI experimental papers. Choose the source that gives the most usable device-level geometry and electrical data, not merely the most famous company name.

Stage 3 - Calibrate TCAD. Establish the baseline and document the physics models. Only after calibration should the provisional ranges be adjusted and frozen.

Stage 4 - Freeze PPA equations. Define power, performance, area and electrical constraints with units and bias conditions.

Stage 5 - DOE and TCAD dataset. Generate an efficient, physically valid sample rather than every possible combination.

Stage 6 - AI surrogate. Compare appropriate surrogate models and report held-out errors. High R-squared alone is not enough; errors at the Pareto region are especially important.

Stage 7 - Multi-objective search. Obtain the Pareto frontier under constraints and inspect which inputs dominate different regions.

Stage 8 - TCAD re-validation. Re-simulate selected Pareto candidates and quantify the discrepancy between surrogate prediction and TCAD.

Stage 9 - Physics interpretation. Explain why the selected geometry/voltage combinations improve one objective and penalize another. This is what turns optimization output into semiconductor research.

Novelty should **not** be claimed as 'new numerical ranges', 'first TCAD + AI GAAFET', 'first ANN prediction' or 'first multi-objective GAA optimization'. Those areas already have prior work.

The defensible hypothesis to investigate is whether an **industry/experiment-informed, calibrated GAA nanosheet model**, combined with **explicit PPA objectives, stacking/operating variables**, efficient surrogate modelling, Pareto analysis and **fresh TCAD validation**, produces new engineering insight.

Final provisional working set: Lg 10-16 nm; WNS 15-35 nm; TNS 4-7 nm; Lsp 3-7 nm; Tsp 8-12 nm provisional; NNS = 2/3/4; EOT approximately 0.7-1.3 nm provisional; VDD 0.5-0.7 V. Doping, work function and materials remain fixed initially to the calibrated reference.

Key references: Xu et al., IEEE TED 2022, DOI 10.1109/TED.2022.3175708; Zhao et al., Micromachines 2024, DOI 10.3390/mi15020218; Pawinanto et al., IJNeAM 2026, DOI 10.58915/ijneam.v19i2.3105. A systematic literature update is required before making final novelty claims.